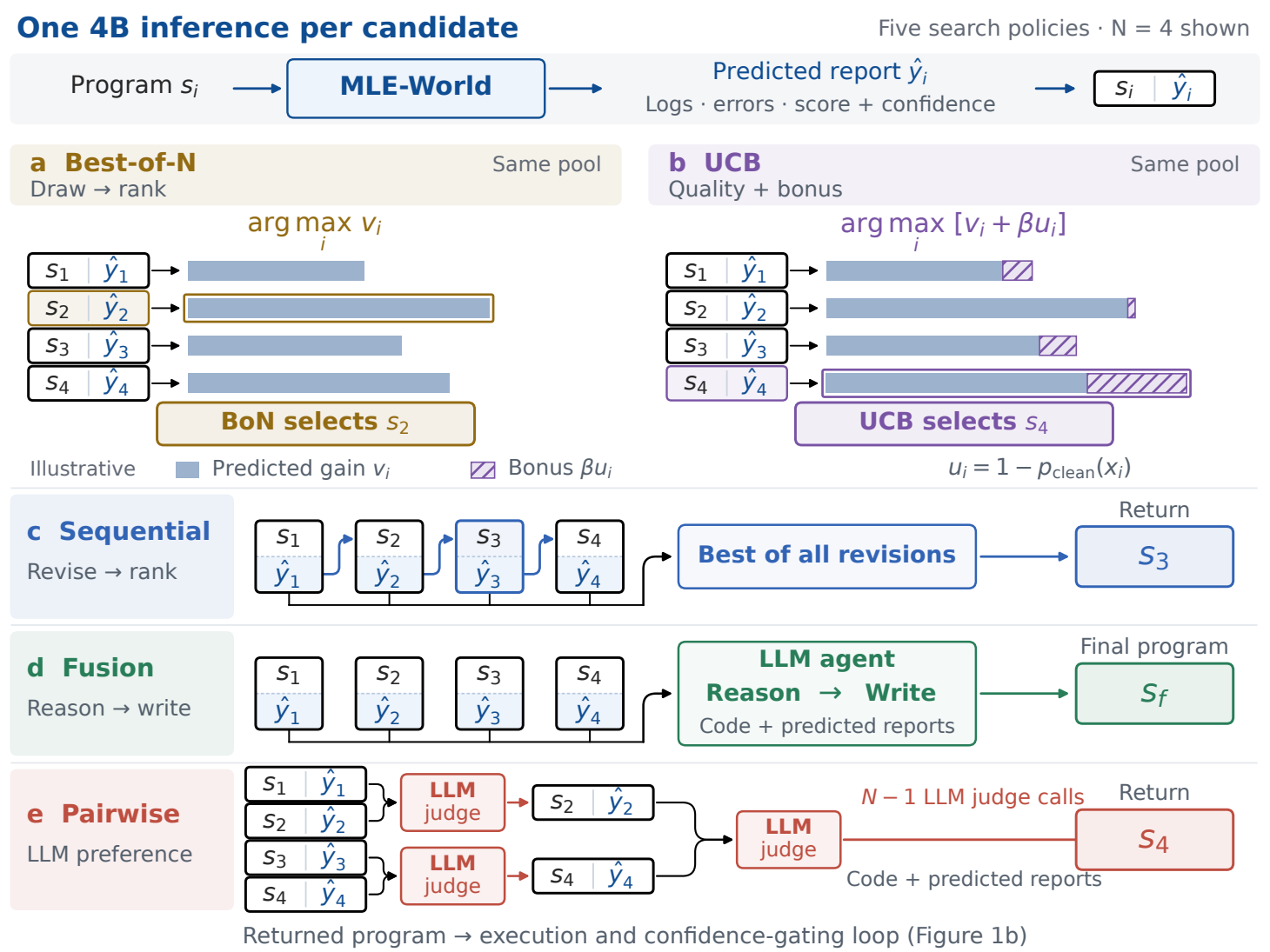
One 4B inference per candidate
Five search policies · N = 4 shown
Program s_i
MLE-World
Predicted report $\hat{y}_i$
Logs · errors · score + confidence
s_i $\hat{y}_i$
a Best-of-N
Draw → rank
Same pool
$\arg\max_i v_i$
s_1 $\hat{y}_1$
s_2 $\hat{y}_2$
s_3 $\hat{y}_3$
s_4 $\hat{y}_4$
BoN selects s_2
b UCB
Quality + bonus
Same pool
$\arg\max_i [v_i + \beta u_i]$
UCB selects s_4
$u_i = 1 - p_{\text{clean}}(x_i)$
Illustrative
Predicted gain v_i
Bonus βu_i
c Sequential
Revise → rank
Best of all revisions
Return
s_3
d Fusion
Reason → write
LLM agent
Reason → Write
Code + predicted reports
Final program
s_f
e Pairwise
LLM preference
LLM judge
LLM judge
LLM judge
$N - 1$ LLM judge calls
Code + predicted reports
Return
s_4
Returned program → execution and confidence-gating loop (Figure 1b)